

i2B2 Admin Help Documentation

Version 1.8.3

Updated 06/2026

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1. Administration & Configuration Plugin Introduction

Now that you have finished installing the i2b2 cells and clients the next step is to setup your users and projects in the i2b2 Administration & Configuration Module. This chapter will contain various sections on how to save data and parameters for the users, projects and hive. There is also a section on the different Authentication Methods supported by the i2b2.

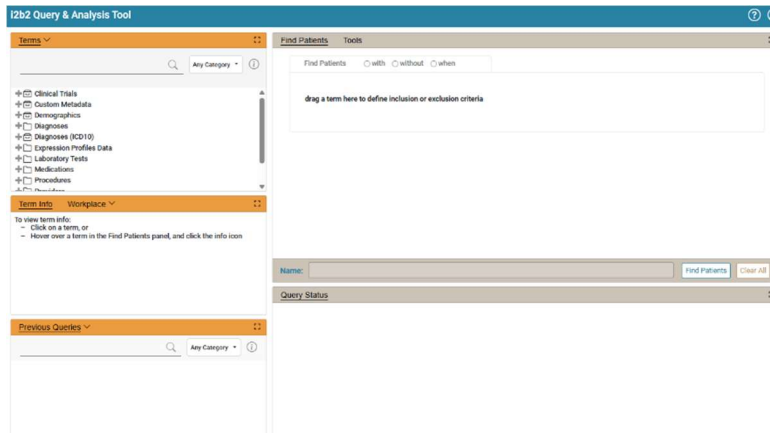
The main goals of the Administration & Configuration module are to help:

1. Manage users and permissions
2. Create projects and manage user access to projects
3. Define HIVE settings

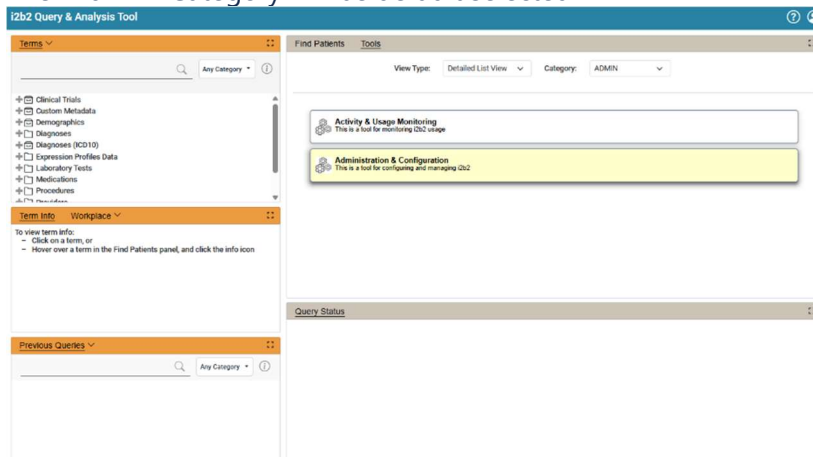
1.1 Accessing Administration & Configuration Module

The i2b2 Administration & Configuration Module can be accessed as a plug in.

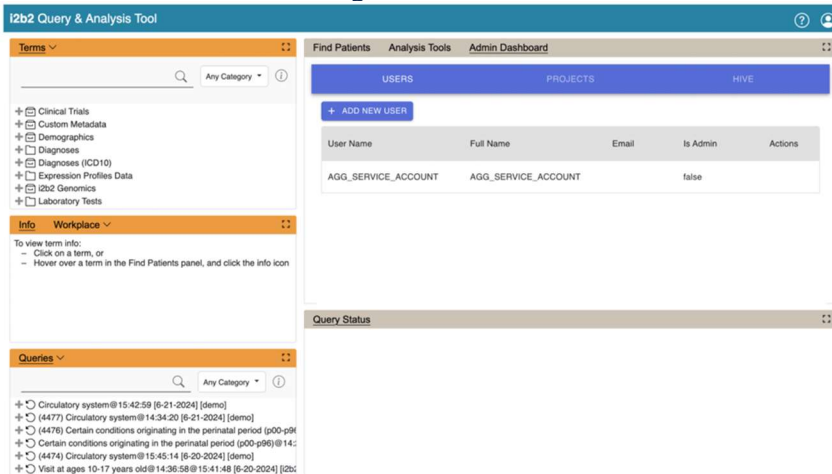
1. Click "Tools"



2. The "Admin" Category will be default selected



3. Click Administration & Configuration



4. This will load the i2b2 Administration & Configuration Module displaying the User tab.

2. Managing i2b2 Users

Everyone who uses the i2b2 has to be setup as an i2b2 user in the i2b2 Administration Module. The management of these users is done under **Users** and the following functions for managing your users are available.

1. Add, edit, and delete user(s) and user's permission setting
2. Add, edit, and delete user parameter(s)

The steps on how to use these features are defined in one of the sections that follow

2.1 User Data

The **User Data** page captures the following general information about a user.

Field	Description	Required
User Name	- A unique id for the user. - It is used when logging into the i2b2 Web Client, Workbench and Admin.	Yes
User Full Name	- The name of the user. - This name will appear in the i2b2 Web Client and Workbench when they log in.	Yes

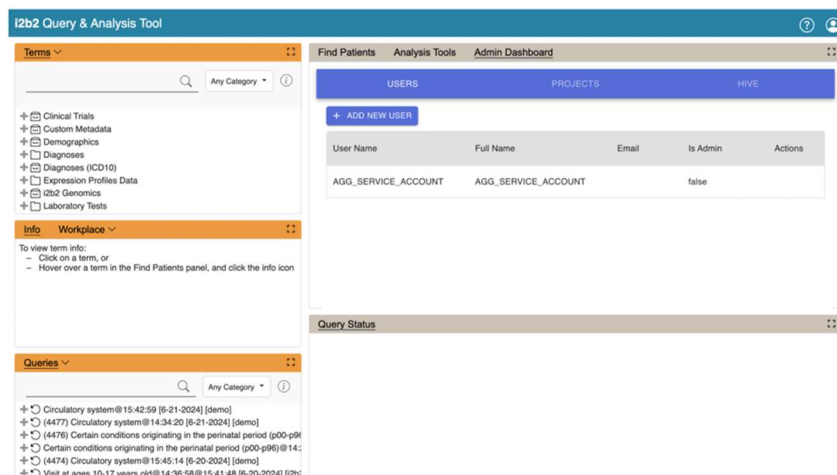
Field	Description	Required
	The name will also appear in the i2b2 Admin under Manage Users in the Navigation panel.	
User Email	At this time, this field is not implemented in the i2b2 core cells and clients.	No
User Password	- The password the user will enter when logging into the i2b2 Web Client, Workbench and Admin. - The password will be encrypted when it is saved to the i2b2 database.	Yes
Is Admin	- A Yes response means the user is an administrator. This means they can log into the i2b2 Admin and will have access to all the cells, projects and users. - A No response means the user is not an administrator. They cannot log into the i2b2 Admin. - The default setting is No	Yes

2.1.1 Add a New User

The following steps outline the process of adding a new user in the i2b2 Administration Module.

Assumption: The following steps assume you are already logged into the i2b2 Administration Module. If you do not know how to login please see the chapter titled *Administration & Configuration Plugin Introduction*.

1. In the Admin Dashboard, click on **Users**. It is the default view. The AGG_SERVICE_ACCOUNT is automatically added through the i2b2 Installation guide.



2. Click "+ADD NEW USER" button. The page to add users will display.

The screenshot shows the 'Admin Dashboard' with tabs for 'Find Patients', 'Analysis Tools', and 'Admin Dashboard'. The 'USERS' tab is active, showing a '+ ADD NEW USER' button and a 'back to All Users' link. The form is divided into 'USER' and 'PARAMETERS (OPTIONAL)' sections. The 'USER' section contains fields for 'User Name *', 'Full Name *', 'Email', 'Password *', 'Verify Password *', and 'Is Admin' (set to 'No'). The 'PARAMETERS (OPTIONAL)' section is empty. At the bottom are 'CANCEL' and 'SAVE' buttons.

3. Enter the information for the user you are adding.

The screenshot shows the 'Add New User' form with the following information entered: 'User Name *' is 'JaneDoe', 'Full Name *' is 'Jane A Doe', 'Email' is empty, 'Password *' is masked with '....', 'Verify Password *' is masked with '....', and 'Is Admin' is 'No'. A green notification banner at the top says 'Saved user JaneDoe'. The 'CANCEL' and 'SAVE' buttons are at the bottom.

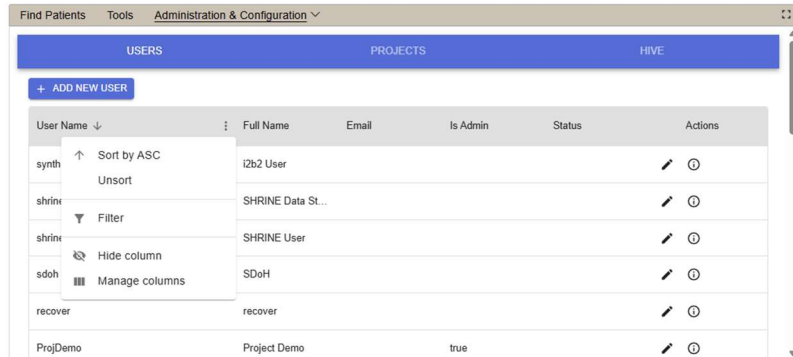
Warning

If **Is Admin** is set to Yes then the user will be able to log into i2b2 Admin Module. This means they are able to access information for ALL users, projects and cells.

4. Click on **Save** to save the user. The user will be added to the list of users on the *Manage* page.

2.1.2 Sort, Filter and Search in User Table

The user details table can be sorted, filtered, or searched by any of the columns.



User Name	Full Name	Email	Is Admin	Status	Actions
synth	i2b2 User				
shrine	SHRINE Data St...				
shrine	SHRINE User				
sdoH	SDoH				
recover	recover				
ProjDemo	Project Demo		true		

- When you hover over the column, three dots will appear on the right end of the column
- Click the three dots to display view options
- Sort allows for ascending or descending sort
- Filter includes standard filter operators to define the filtered value
- To Search for a specific user, select filter and filter by the specific User Name or Full Name

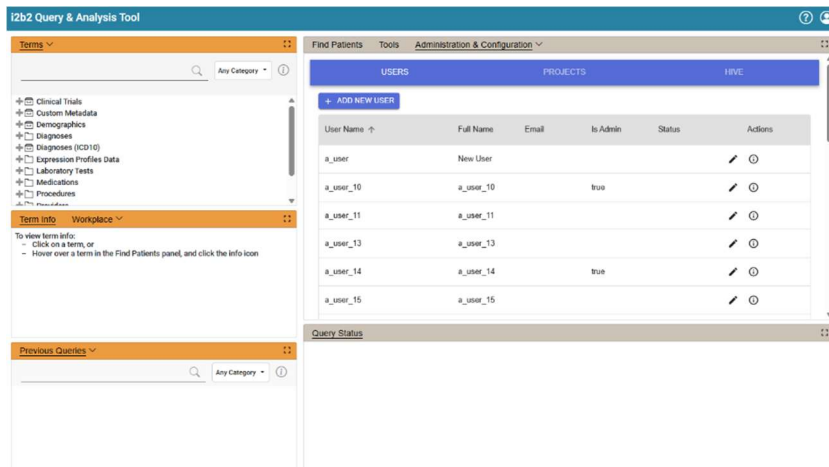
Columns	Operator	Value
X User Name	contains	Filter value

2.1.3 Edit a User in i2b2

The following steps outline the process of editing a user in the i2b2 Administration Module.

Assumption: The following steps assume you are already logged into the i2b2 Administration Module. If you do not know how to login please see the chapter titled *Administration & Configuration Plugin Introduction*.

1. In the Admin Dashboard, click on **Users**. It is the default view.



2. Click the **pencil icon** in the row of the **name of the user** you want to edit.
3. The *User* page will display.

4. Make the changes to the user and click on **Save** to save the changes.

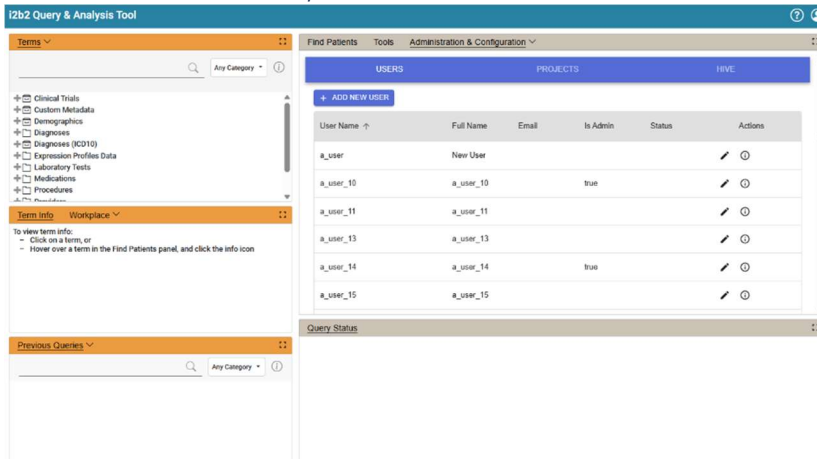
2.1.4 Delete a User in i2b2

The following steps outline the process of deleting a user in the i2b2 Administration Module.

Assumption: The following steps assume you are already logged into the i2b2 Administration Module. If you do not know how to login please see the chapter titled *Administration &*

Configuration Plugin Introduction.

1. In the Admin Dashboard, click on **Users**. It is the default view.



2. Click the **pencil icon** in the row of the **name of the user** you want to edit.
3. The **User** page will display.

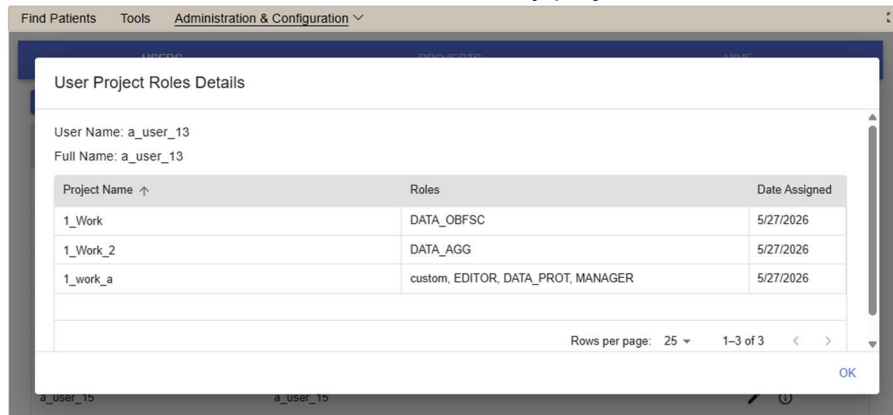
The screenshot shows the 'User' edit page in the I2B2 Query & Analysis Tool Admin Dashboard. The page has a header with 'Find Patients', 'Tools', and 'Administration & Configuration'. The 'Administration & Configuration' dropdown is active, showing 'USERS', 'PROJECTS', and 'HIVE'. The 'USERS' tab is selected. Below the navigation bar, there is a 'back to All Users' link. The page is divided into two sections: 'USER' and 'PARAMETERS (OPTIONAL)'. The 'USER' section contains form fields for 'User Name', 'Full Name', 'Email', 'Is Admin', and 'Authentication Config'. The 'PARAMETERS (OPTIONAL)' section is currently empty. At the bottom of the page, there are three buttons: 'DELETE USER', 'CANCEL', and 'SAVE'.

4. Click the **Delete User** button.
5. Confirm the action. You will be returned to the **Users** page and the user will be removed from the list of users on the page.

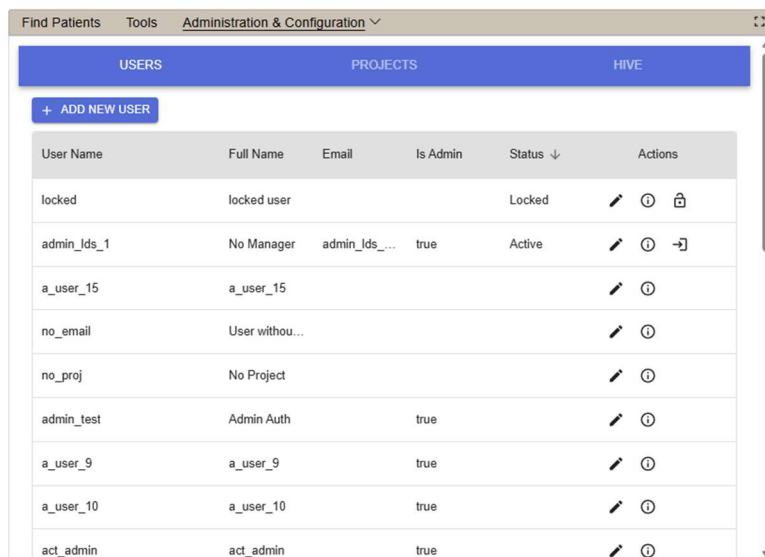
2.2 User Table Actions

Under the Actions heading in the list of users, an admin may see options for up to 4 actions depending on each user's status.

1. A Pencil icon, to edit a user as described above
2. An info icon, to show user details by project



3. An exit, or logout, icon, to terminate a user's active session
4. An unlock icon, to unlock a user who has been locked out.



2.3 User Parameters

Parameters entered on a user are known as User Parameters and are specific to the user. The user parameter only affects the user in which it is entered on; it does not affect other users and it is not specific to any one project.

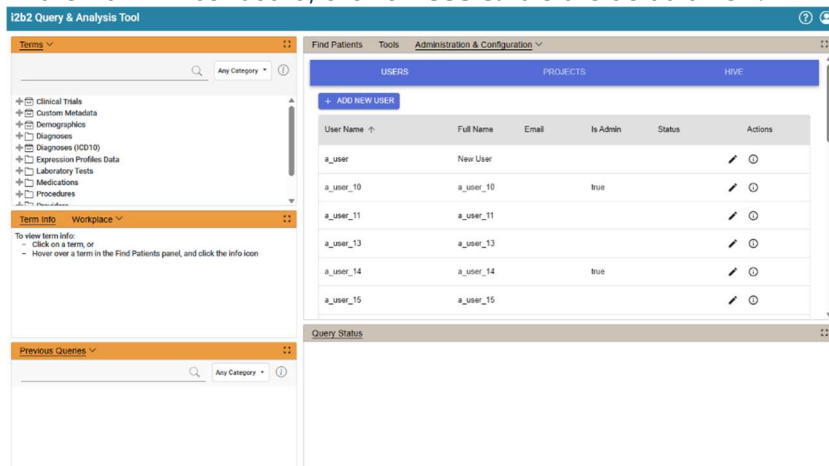
User Parameters	• Specific to the user in which it is defined. • Does not affect other users. • Is not specific to any one project.
-----------------	--

2.3.1 Add a User Parameter

The following steps outline the process of adding a parameter to a user from within the i2b2 Administration Module.

Assumption: The following steps assume you are already logged into the i2b2 Administration Module. If you do not know how to login please see the chapter titled *Administration & Configuration Plugin Introduction*.

1. In the Admin Dashboard, click on **Users**. It is the default view.



2. Click the **pencil icon** in the row of **name of the user** you want to add a parameter to.
3. Click on the **Parameters (optional)** option that displays next to the User's tab.

The screenshot shows the 'Admin Dashboard' with the 'USERS' tab selected. Below the tab bar is a '+ ADD NEW USER' button and a 'back to All Users' link. There are two tabs: 'USER' and 'PARAMETERS (OPTIONAL)'. The 'USER' tab is active, showing a form with the following fields: 'User Name *' (JaneDoe), 'Full Name *' (Jane A Doe), 'Email' (empty), 'Password *' (masked with dots), 'Verify Password *' (masked with dots), and 'Is Admin' (No). There are 'CANCEL' and 'SAVE' buttons at the bottom.

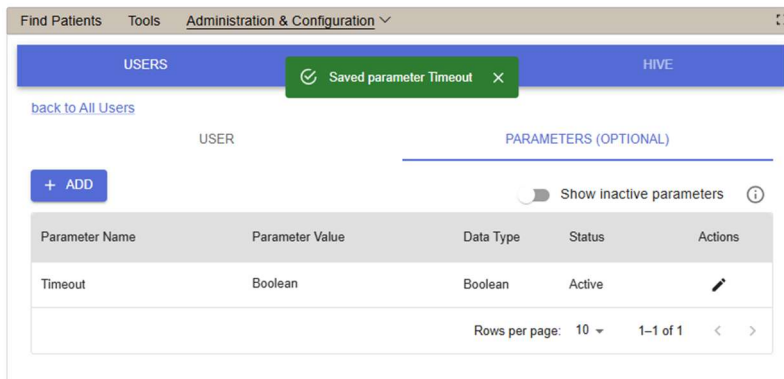
4. The **Parameters List** page will display.

The screenshot shows the 'Admin Dashboard' with tabs for 'Find Patients', 'Analysis Tools', and 'Admin Dashboard'. Below the tabs is a navigation bar with 'USERS', 'PROJECTS', and 'HIVE'. A link 'back to All Users' is visible. The main content area is titled 'USER' and 'PARAMETERS (OPTIONAL)'. There is a '+ ADD' button. Below it is a table with columns: Name, Value, Data Type, and Actions. The table is empty, showing 'No rows'. At the bottom right, it says 'Rows per page: 5' and '0-0 of 0'.

5. Click on the **Add button**. A new row will display.

This screenshot is similar to the previous one, but the '+ ADD' button has been clicked, and a new row has been added to the table. The 'Data Type' dropdown menu is open, showing options: Text, Numeric, Date, Integer, Boolean, Reference Binary, RTF, Word, Excel, and XML. The 'Text' option is selected. The table now shows '1 of 1' row.

- Enter the name of the parameter, the value for the parameter and the data type for the parameter.
- Click on **Save** to save the new parameter.
- The *Parameters List* page will display with the new parameter.

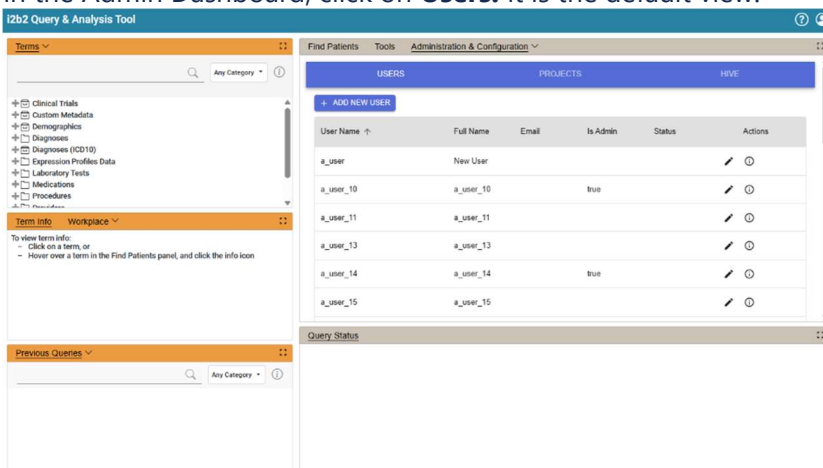


2.3.2 Edit a User Parameter

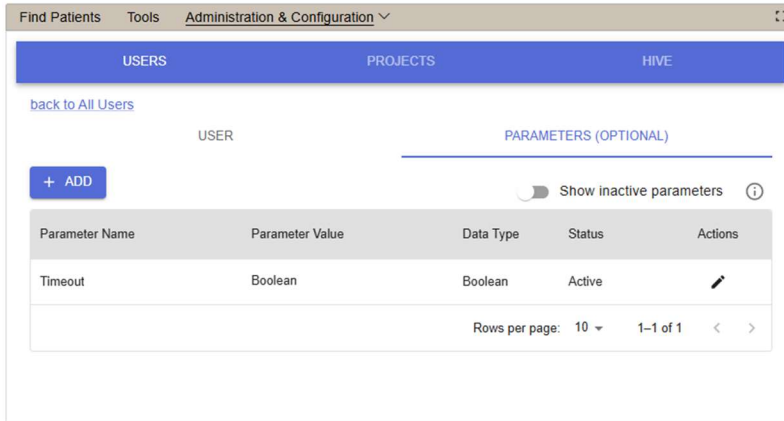
The following steps outline the process of editing a user parameter in the i2b2 Administration Module.

Assumption: The following steps assume you are already logged into the i2b2 Administration Module. If you do not know how to login please see the chapter titled *Administration & Configuration Plugin Introduction*.

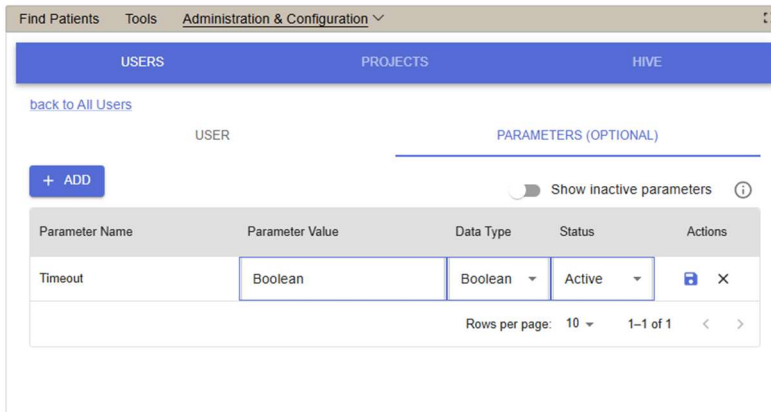
- In the Admin Dashboard, click on **Users**. It is the default view.



- Click the **pencil icon** in the row of the **name of the user** you want to add a parameter to.
- Click on the **Parameters (optional)** option that displays next to the user's tab.



4. The **Parameters List** page will display. A list of parameters entered for that user will display.
5. Click the **pencil icon** in the row of the **name of the parameter** you want to edit.
6. The *Edit Parameter* fields will display for that row.



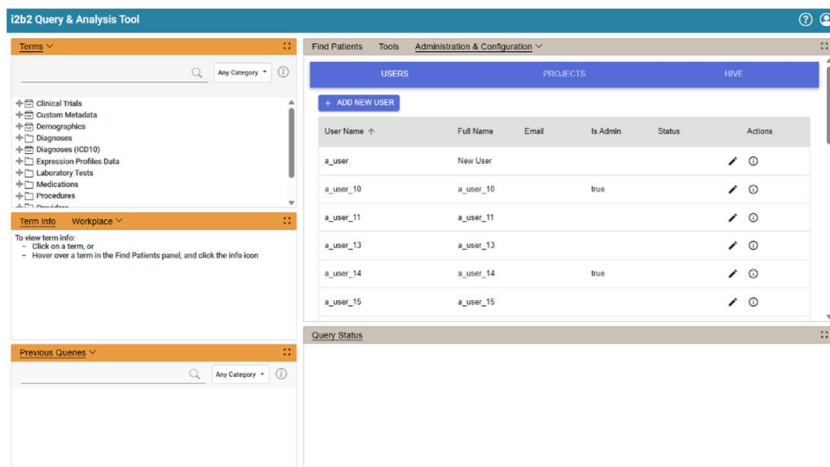
7. Make the changes to the parameter and click on the **Save** icon to save the changes.

2.3.3 Delete a User Parameter

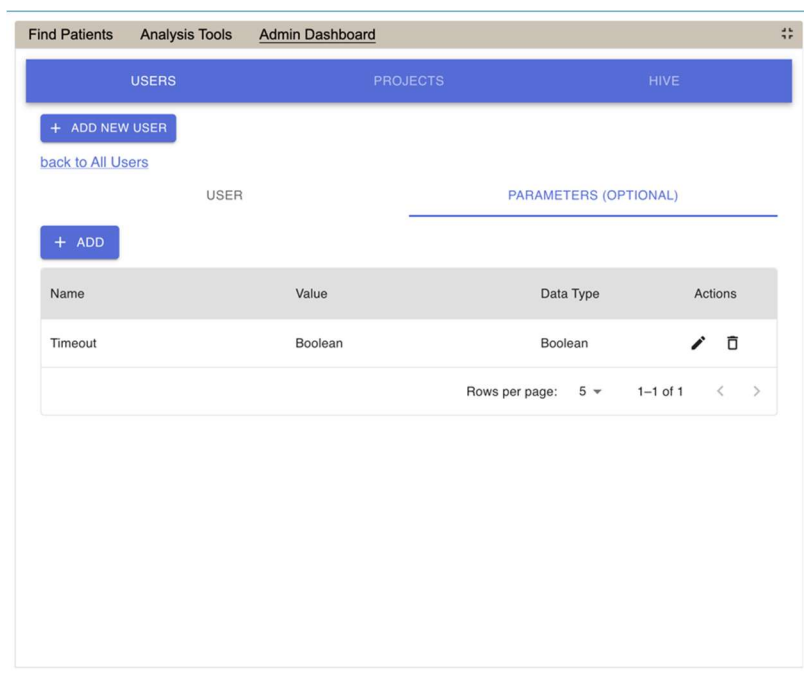
The following steps outline the process of deleting a user parameter in the i2b2 Administration Module.

Assumption: The following steps assume you are already logged into the i2b2 Administration Module. If you do not know how to login please see the chapter titled *Administration & Configuration Plugin Introduction*.

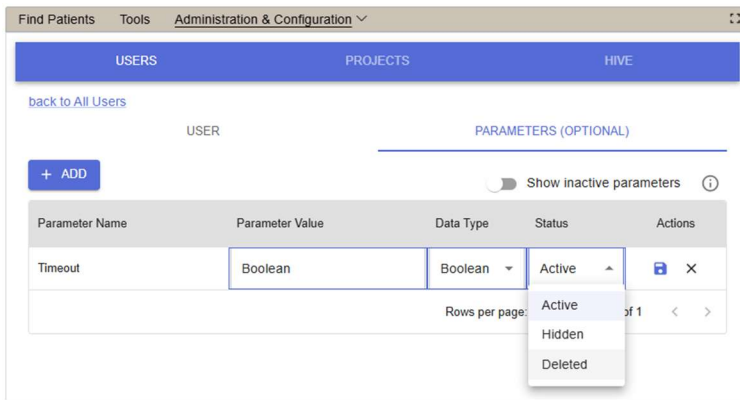
1. In the Admin Dashboard, click on **Users**. It is the default view.



2. Click the **pencil icon** in the row of the **name of the user** you want to delete a parameter for.
3. Click on the **Parameters (optional)** option that displays next to the user's tab.



4. The **Parameters List** page will display. A list of parameters entered for that user will display.
5. Click the **pencil icon** in the row of the **name of the parameter** you want to delete.
6. The *Edit Parameter* fields will display for that row.
7. Select **Deleted** from the **Status field** dropdown in the row of the **name of the parameter** you want to delete.



8. Click the **Save** icon and confirm the action. The parameter will be removed from the list of parameters for that user.

3. Managing i2b2 Projects

After a user is setup in the i2b2 Administration Module they need to be given access to one or more projects before they can log into the i2b2 Web Client or Workbench. The management of your project data is done in the i2b2 Administration Module within **Projects**. The following functions for managing the projects are available.

Project Data

1. Add, edit, and delete project(s)
2. Add, edit, and delete project parameter(s)
3. Add, edit data sources

Project Users

1. Add and delete project user(s)
2. Add and edit project user roles
3. Add, edit, and delete project user parameter(s)

3.1 Project Data

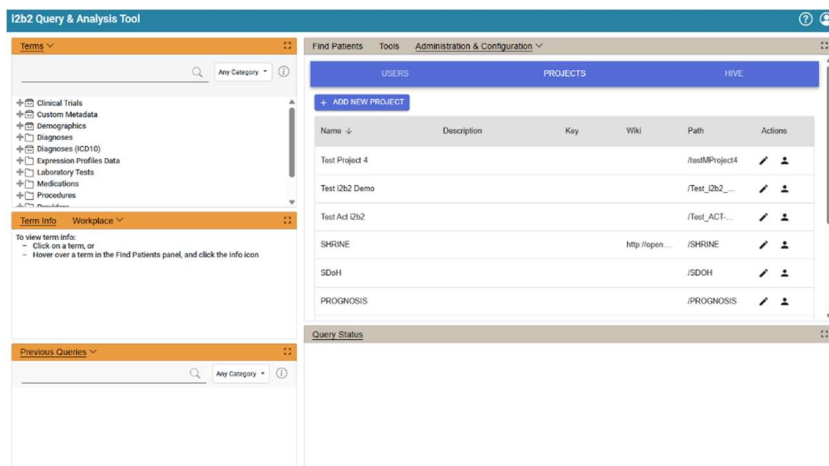
The **Project Data** page captures the following general information about a project.

Field	Description	Required
Project ID	A unique identifier for the project.	Yes
Project Name	The name of the project.	Yes

Field	Description	Required
	Visible to users in the i2b2 Web Client and Workbench	
Project Wiki	Currently not implemented in the i2b2 Web Client or Workbench.	No
Project Key	Enter the encryption key that is used during the encryption / decryption process.	No
Project Description	A brief description of the project. At this time it is only visible in the i2b2 Admin.	No
Project Path		Yes

3.1.1 Add a New Project

1. In the Admin Dashboard, click on **Projects**.



2. Click "+ADD NEW PROJECT" button. The page to add **project-details** will display.

Find Patients Analysis Tools Admin Dashboard

USERS PROJECTS HIVE

back to All Projects

1 Project Details 2 Parameters Optional! 3 Data Sources 4 User Associations

New Project - Details

ID *

Name *

Path *

Wiki

Key

Description

EXIT SAVE AND CONTINUE

- Once details have been added, click **Save and Continue**.
- To add optional Project specific Parameters. See 3.2 Project Parameters.
- To add Data Sources. See 3.3 Project Data Source.
- See 3.4 to Add Users to the Project. See 3.5 Add User's Project Roles, and see 3.6 User's Project Parameters.

3.1.2 Edit a New Project

- In the Admin Dashboard, click on **Projects**.

Find Patients Tools Administration & Configuration

USERS PROJECTS HIVE

+ ADD NEW PROJECT

Name ↓	Description	Key	Wiki	Path	Actions
Test Project 4				/testMProject4	✎
Test 12b2 Demo				/Test_12b2_...	✎
Test Act 12b2				/Test_ACT...	✎
SHRINE			http://open...	/SHRINE	✎
SDOH				/SDOH	✎
PROGNOSIS				/PROGNOSIS	✎

Query Status

- Click the **pencil icon** in the row of the **name of the project** you want to edit.

Find Patients Analysis Tools Admin Dashboard

USERS PROJECTS HIVE

[back to All Projects](#)

1 Project Details 2 Parameters Optional 3 Data Sources 4 User Associations

New Project - Details

ID *

Name *

Path *

Wiki

Key

Description

EXIT SAVE AND CONTINUE

3. Edit details and click Save and Continue

3.1.3 Delete a New Project

1. In the Admin Dashboard, click on **Projects**.

12b2 Query & Analysis Tool

Find Patients Tools Administration & Configuration

USERS PROJECTS HIVE

+ ADD NEW PROJECT

Name	Description	Key	Wiki	Path	Actions
Test Project 4				/testMPProject4	
Test 12b2 Demo				/Test_12b2_...	
Test Act 12b2				/Test_ACT...	
SHRINE			http://open...	/SHRINE	
SDOH				/SDOH	
PROGNOSIS				/PROGNOSIS	

Query Status

2. Click the **pencil icon** in the row of the **name of the project** you want to edit.
3. The *Project Details* page will display.

Find Patients Tools Administration & Configuration

USERS PROJECTS HIVE

[back to All Projects](#)

1 Project Details 2 Parameters Optional 3 Data Sources 4 User Associations

1_Work - Details

ID *

1_work_1

Name *

1_Work

Path *

/1_work_1

EXIT DELETE PROJECT SAVE CONTINUE

- Click the **Delete Project** button.
- Confirm the action. You will be returned to the **Projects** page and the project will be removed from the list of users on the page.

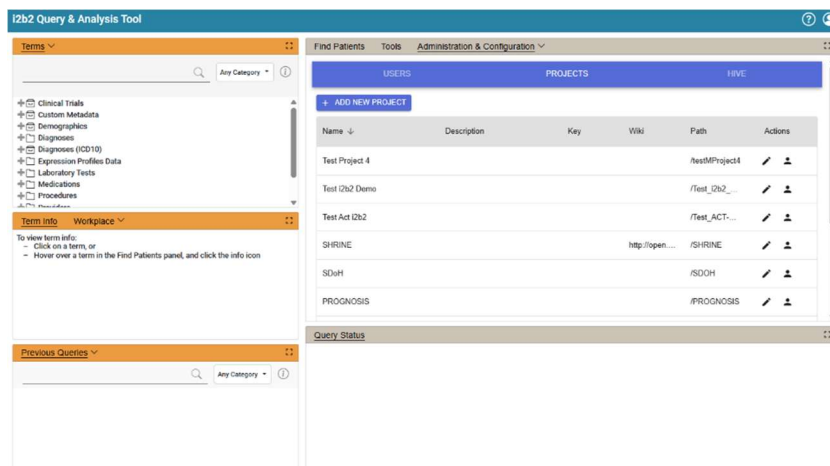
3.2 Project Parameter

Parameters entered on a project are known as **Project Parameters** and are *specific to the project*. The project parameter only effects the project for which it is entered; it does not affect the other projects in the hive.

Project Parameters	- Specific to the project in which it is defined. - Does not affect other projects. - Will affect all users who have access to the project.
--------------------	--

3.2.1 Add a Parameter to a Project

- In the Admin Dashboard, click on **Projects**.



- Click the **pencil icon** in the row of the **name of the project** you want to edit.

Find Patients Analysis Tools Admin Dashboard

USERS PROJECTS HIVE

back to All Projects

1 Project Details 2 Parameters Optional 3 Data Sources 4 User Associations

New Project - Details

ID *

Name *

Path *

Wiki

Key

Description

EXIT SAVE AND CONTINUE

- Click on the **Parameters (optional)** option that displays next to the project details tab.

Find Patients Analysis Tools Admin Dashboard

USERS PROJECTS HIVE

back to All Projects

✓ Project Details 2 Parameters Optional 3 Data Sources 4 User Associations

safari - Parameters

+ ADD

Name	Value	Data Type	Actions
No rows			

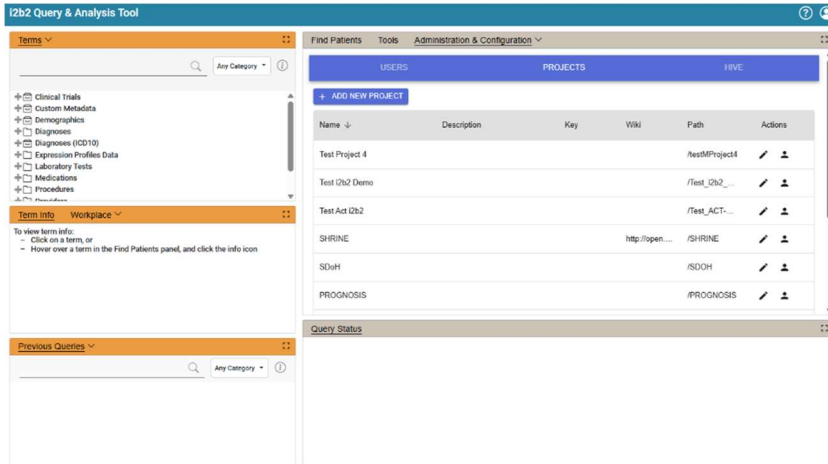
Rows per page: 5 0-0 of 0

EXIT PREVIOUS SAVE AND CONTINUE

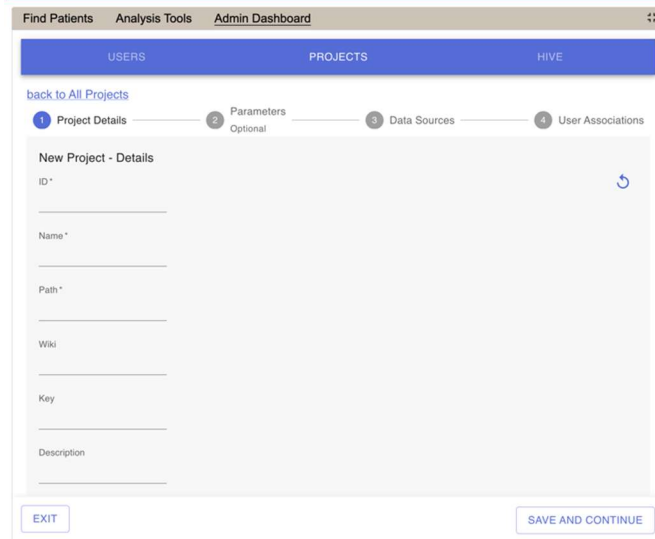
- The **Parameters List** page for the project will display. A list of parameters entered for that project will display.
- Click "Add" to add a new parameter. A new row will display.
- Enter the name of the parameter, the value for the parameter and the data type for the parameter.
- Click the **Save** icon to save the new parameter.

3.2.2 Edit a Parameter for a Project

1. In the Admin Dashboard, click on **Projects**.



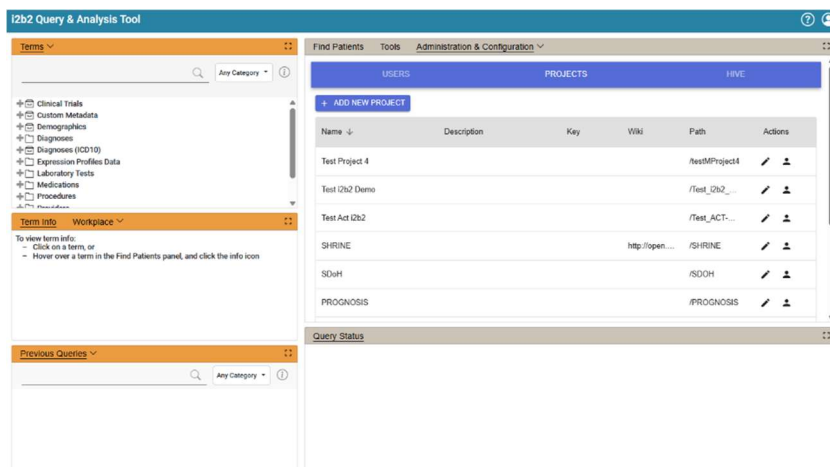
2. Click the **pencil icon** in the row of the **name of the project** you want to edit.



3. Click on the **Parameters (optional)** option that displays next to the project details tab.
4. The **Parameters List** page will display. A list of parameters entered for that project will display.
5. Click the **pencil icon** in the row of the **name of the parameter** you want to edit.
6. The *Edit Parameter* fields will display for that row.
7. Make the changes to the parameter and click on the **Save** icon to save the changes.

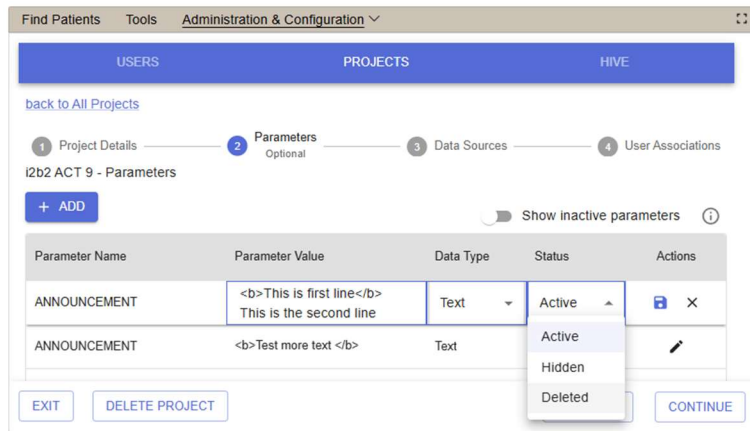
3.2.3 Delete a Parameter for a Project

- 4 In the Admin Dashboard, click on **Projects**.



- 5 Click the **pencil icon** in the row of the **name of the project** you want to edit.

- 6 Click on the **Parameters (optional)** option that displays next to the project details tab.
- 7 The **Parameters List** page will display. A list of parameters entered for that project will display.
- 8 Click the **pencil icon** in the row of the **name of the parameter** you want to delete.
- 9 The *Edit Parameter* fields will display for that row.
- 10 Select **Deleted** from the **Status field** dropdown in the row of the **name of the parameter** you want to delete.

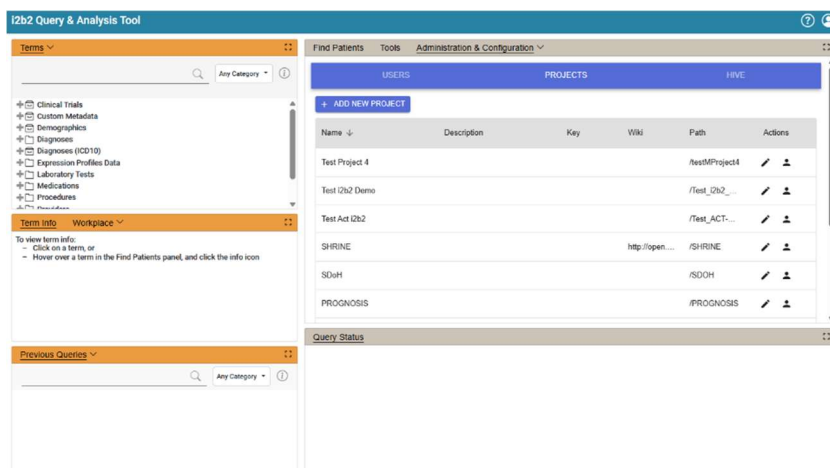


- 11 Confirm the action. The parameter will be removed from the list of project parameters on the *Project* parameters page, but can still be displayed by toggling on “Show inactive parameters”.

3.3 Project Data Source

3.3.1 Add a Data Source to a Project

1. In the Admin Dashboard, click on **Projects**.



2. Click the **pencil icon** in the row of the **name of the project** you want to edit.

Find Patients Analysis Tools Admin Dashboard

USERS PROJECTS HIVE

back to All Projects

1 Project Details 2 Parameters Optional 3 Data Sources 4 User Associations

New Project - Details

ID *

Name *

Path *

Wiki

Key

Description

EXIT SAVE AND CONTINUE

3. Click **Data Sources** for the **Data Source** page to display. (NOTE: when creating a new project, you may need to click Save and Continue on the **Project Details** and **Parameters** pages in order to add data sources.)

Find Patients Analysis Tools Admin Dashboard

USERS PROJECTS HIVE

back to All Projects

1 Project Details 2 Parameters Optional 3 Data Sources 4 User Associations

I2B2UI-814 - Data Source Details

CRC Details

Database Server *

Name *

Project Path *

/test

JNDI Data Source *

java:/

Database Schema *

ONT Details

Database Server *

Name *

Project Path *

/test

JNDI Data Source *

java:/

Database Schema *

WORK Details

Database Server *

Name *

Project Path *

/test

JNDI Data Source *

java:/

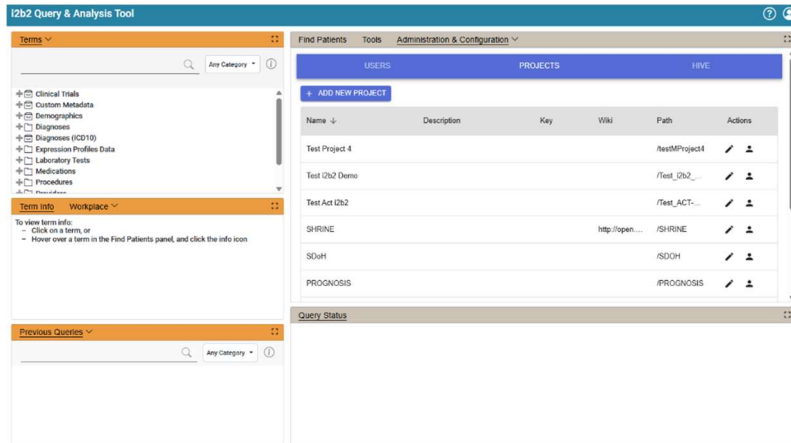
Database Schema *

EXIT PREVIOUS SAVE AND CONTINUE

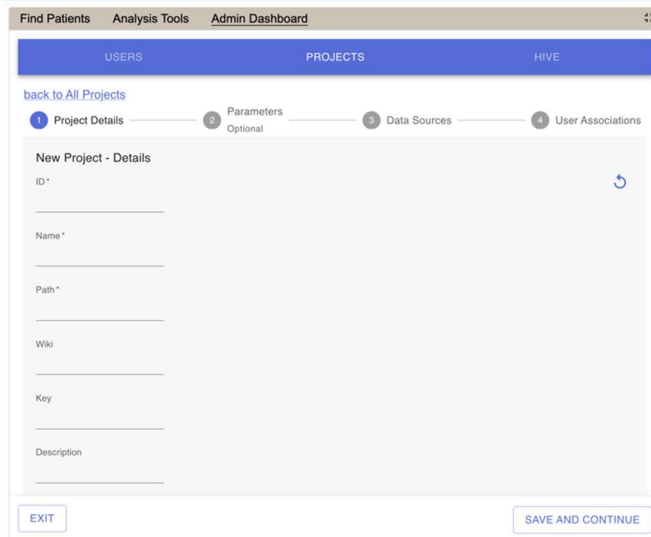
4. Enter details for the CRC, ONT, and WORK data sources. Click Save and Continue.
 - a. NOTE for instructions of details needed for data sources, refer to this page on the i2b2 wiki:
<https://community.i2b2.org/wiki/display/getstarted/2.+Data+Source+Configuration>

3.3.2 Edit a Data Source for a Project

1. In the Admin Dashboard, click on **Projects**.



2. Click the **pencil icon** in the row of the **name of the project** you want to edit.



3. Click the **Data Sources** for the **Data Source** page to display.

The screenshot shows the 'Admin Dashboard' with a navigation bar containing 'Find Patients', 'Analysis Tools', and 'Admin Dashboard'. Below this is a tabbed interface with 'USERS', 'PROJECTS', and 'HIVE'. The 'PROJECTS' tab is active, showing a progress bar with four steps: 'Project Details' (checked), 'Parameters' (Optional, checked), 'Data Sources' (active, highlighted with a blue circle and number 3), and 'User Associations' (highlighted with a blue circle and number 4). A link 'back to All Projects' is visible. The main content area is titled 'I2B2UI-814 - Data Source Details' and contains three side-by-side panels: 'CRC Details', 'ONT Details', and 'WORK Details'. Each panel has a 'Database Server' dropdown, a 'Name' text field, a 'Project Path' text field (with '/test' as a hint), a 'JNDI Data Source' text field (with 'java:/' as a hint), and a 'Database Schema' text field. At the bottom of the form are three buttons: 'EXIT', 'PREVIOUS', and 'SAVE AND CONTINUE'.

4. Edit the details for the CRC, ONT, and WORK data sources. Click Save and Continue.

3.4 Project Users

Each user will have at least two roles per *user_id* and *project_id* association. These two roles can be further defined as a **Data Protection role** and a **Hive Management role**.

The data protection role establishes the detail of data the user can see while the hive management role defines the level of functionality the user has in a project. The following tables summarize the roles in a hierarchical order of least to most access.

Data Protection Roles

Role	Access Description
DATA_OBFSC	OBFSC = Obfuscated - The user can see aggregated results that are obfuscated (example: patient count). - The user is limited on the number of times they can run the same query within a specified time period. If the user exceeds the maximum number of times then their account will be locked and only the Admin user can unlock it.
DATA_AGG	AGG = Aggregated The user can see aggregated results like the patient count.

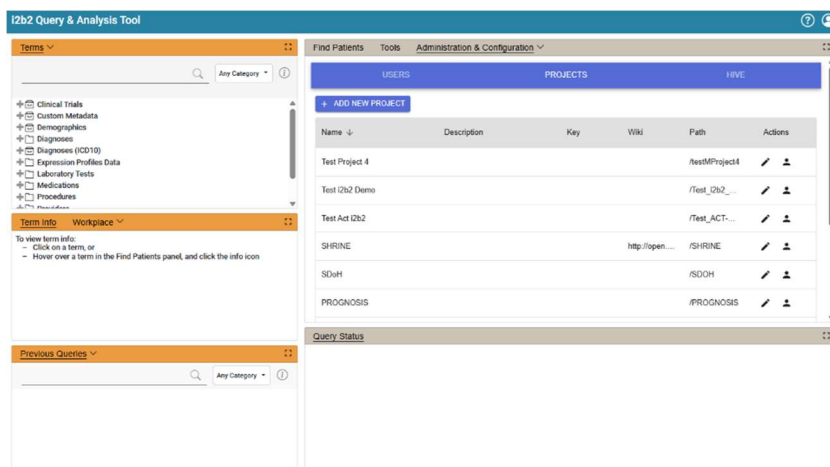
Role	Access Description
	The results are not obfuscated and the user is not limited to the number of times they can run the same query.
DATA_LDS	LDS = Limited Data Set - The user can see all fields except for those that are encrypted. - An example of an encrypted field is the blob fields in the fact and dimension tables.
DATA_DEID	DEID = De-identified Data - The user can see all fields including those that are encrypted. - An example of an encrypted field is the blob fields in the fact and dimension tables.
DATA_PROT	PROT = Protected The user can see all data, including the identified data that resides in the Identity Management Cell.

Hive Management Roles

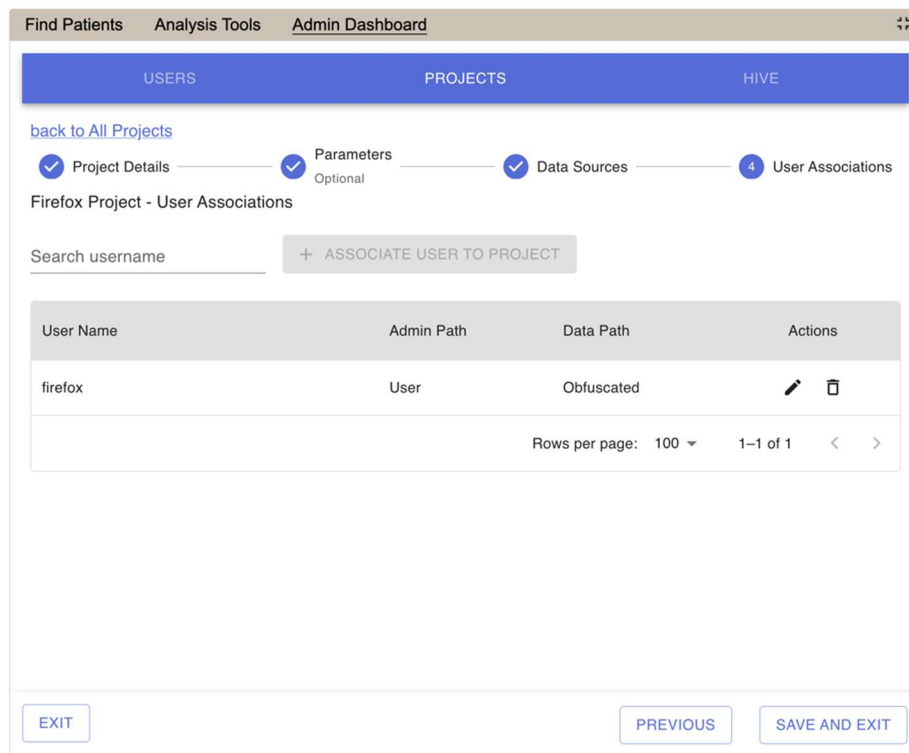
Role	Access Description
USER	Can create queries and access them if he / she is the owner of the query.
MANAGER	Can create queries as well as access queries created by different users within the project.

3.4.1 Define a Project User Roles

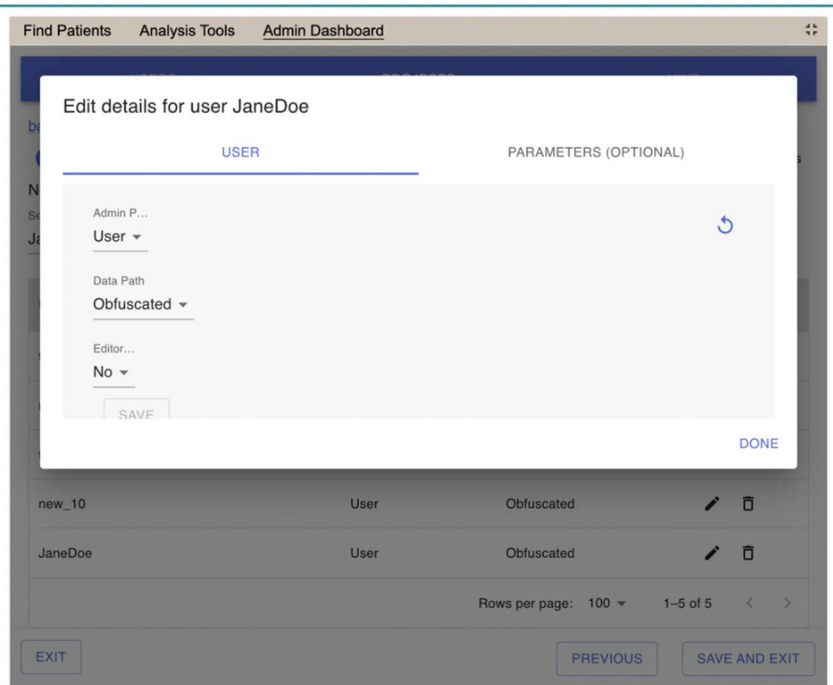
1. In the Admin Dashboard, click on **Projects**.



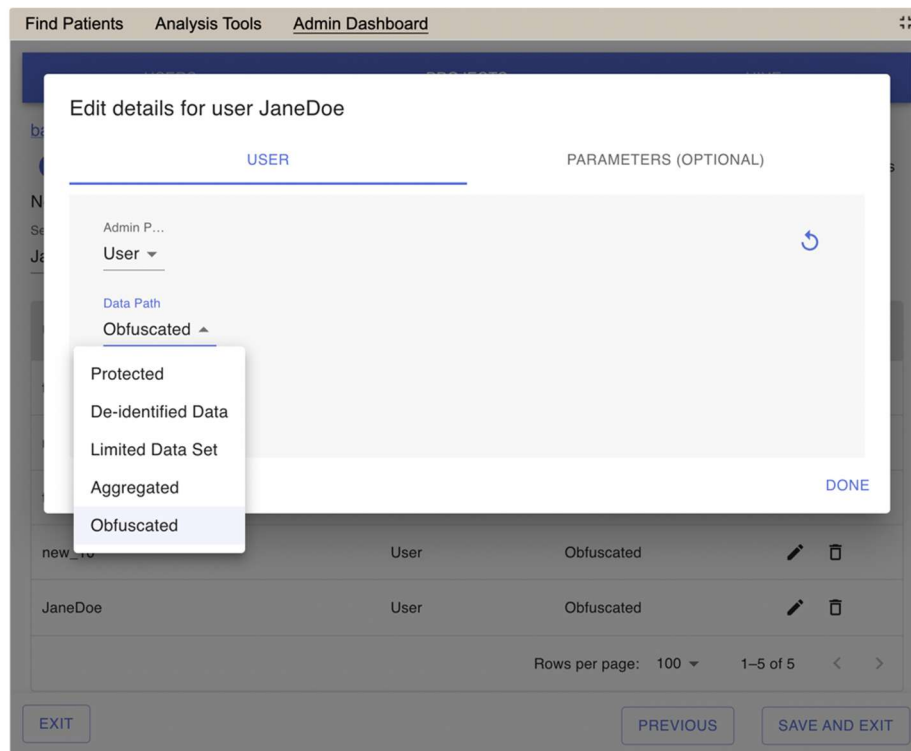
2. Click the **person icon** in the row of the **name of the project** you want to add users to. (The person icon takes you directly to the **User Associations** page of project details interface.)



3. On the **User Association** page for a given project, search by the username.
4. Once a username is identified and selected, click the **Associate User to Project** button.
5. The user will be added to the project, and the **details for user** page open to modify their user roles for the project.



6. Select the **User Role**, and **Data Role**. Custom roles can also be added.



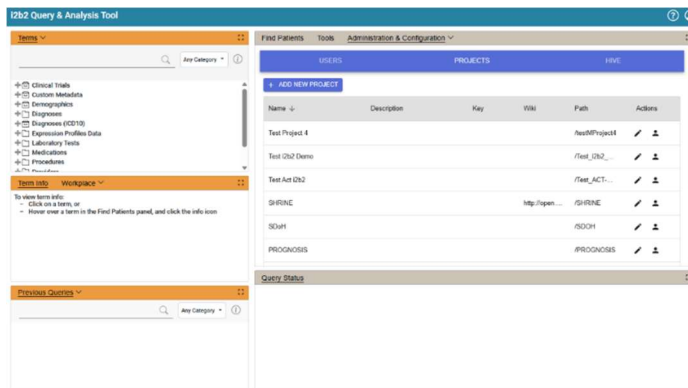
7. Click Save.

3.4.2 Project User Parameters

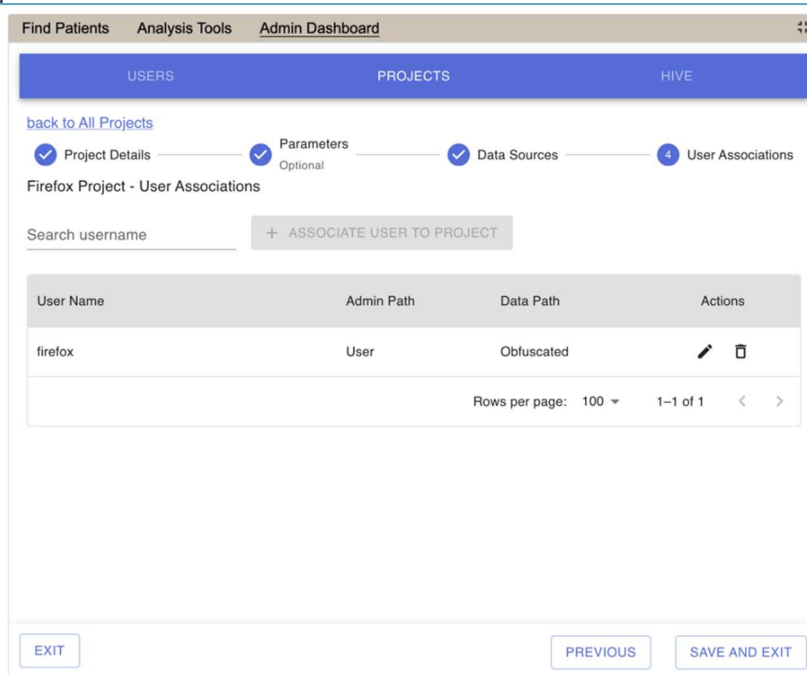
Project User Parameters

- Specific to the user **and** the project in which it is defined.
- Does not affect other users within the project.
- Does not affect other projects the user has access to.

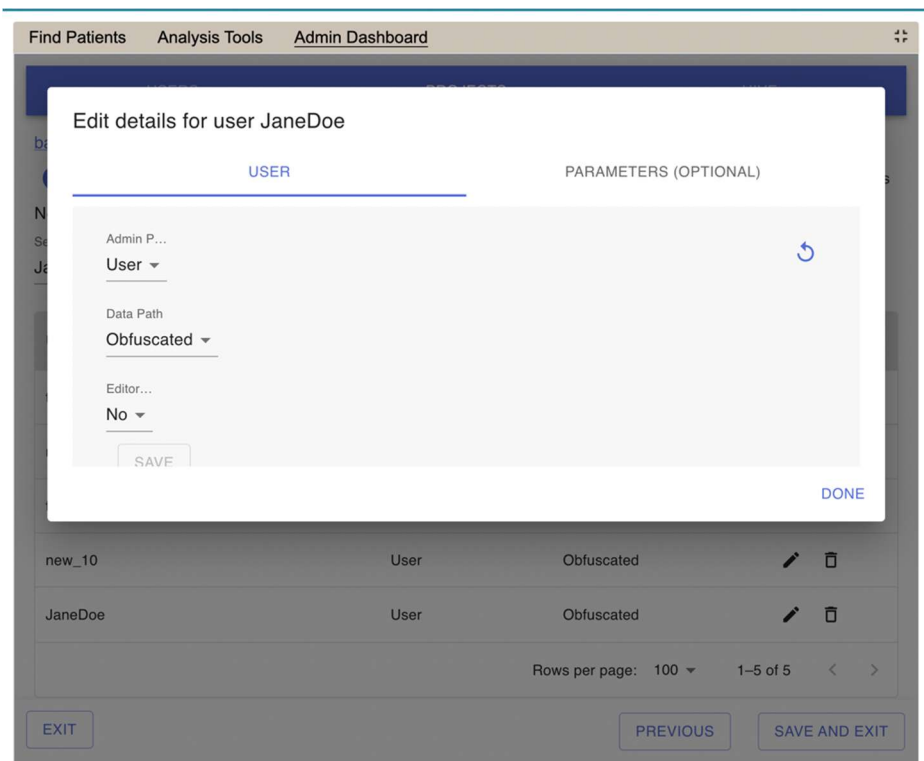
1. In the Admin Dashboard, click on **Projects**.



2. Click the **person icon** in the row of the **name of the project** you want to add project users parameters to.



3. On the **User Associations** page for a given project, click the **pencil icon** next to a user to modify their user role for the project.



4. Select the Parameters tab
5. Click the **Add** button to add new Project-User parameters, or the **pencil icon** to edit or delete existing Project-User parameters.
6. Click done to Save.

4. Managing i2b2 Hive Data

The i2b2 hive and associated web services are the infrastructure used to integrate a collection of cells. The management of your Hive data is done in the i2b2 Administration Module within **Manage Hive**. The following functions for managing the hive are available.

1. Edit hive data (*Domain ID, Help URL, Domain Name, and Environment*)
2. Add, edit, and delete hive (*global*) parameter(s)

The steps on how to use these features are defined in the sections that follow.

4.1 Hive Data

The Hive Data page captures the following general information about a hive.

Field	Description	Required
Help URL	A link to an external website that may contain additional information on using the i2b2 in your environment. This link is only used in the i2b2 Workbench and is not available from within the i2b2 Webclient. <i>(in the workbench heading there is a button labeled i2b2; clicking on this button will launch the website entered at this field.)</i>	No
Domain Name (View Only)	The name of the domain in which this hive resides. This is the same name that is entered in your Web Client and Admin config files as well as the Workbench properties file.	Yes
Environment	The type of environment this hive resides. The options are: Development Production Test	Yes

Note

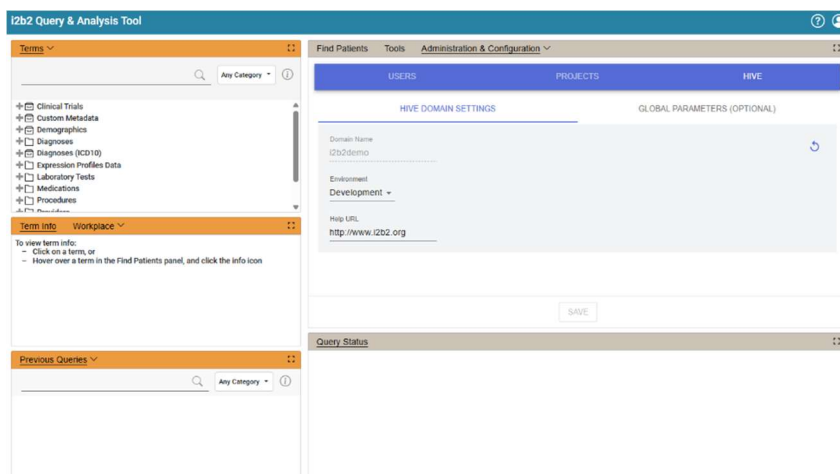
The hive is created when you are installing the i2b2 server. You cannot add or remove a hive from within i2b2 Admin; however you can edit the hive's data.

4.1.1 Edit Hive Data

The following steps outline the process of editing the Hive Data in the i2b2 Administration Module.

Assumption: The following steps assume you are already logged into the i2b2 Administration Module. If you do not know how to login please see the chapter titled *Administration & Configuration Plugin Introduction*.

1. In the Admin Dashboard, click on **Hive**.



2. Click on **Hive Domain Settings**. This is the default view.

The screenshot shows the i2b2 Admin Dashboard. At the top, there is a navigation bar with three tabs: 'Find Patients', 'Analysis Tools', and 'Admin Dashboard'. Below this is a secondary navigation bar with three tabs: 'USERS', 'PROJECTS', and 'HIVE'. The 'HIVE' tab is selected. The main content area is titled 'HIVE DOMAIN SETTINGS' and contains a form with the following fields: 'Domain Name' (with the value 'i2b2ui-test'), 'Environment' (with a dropdown menu showing 'Development'), and 'Help URL' (with the value 'http://www.i2b2.org'). A 'SAVE' button is located at the bottom right of the form.

3. Make the changes to the Hive Data and click on **Save** to save the changes.

4.2 Hive Parameters

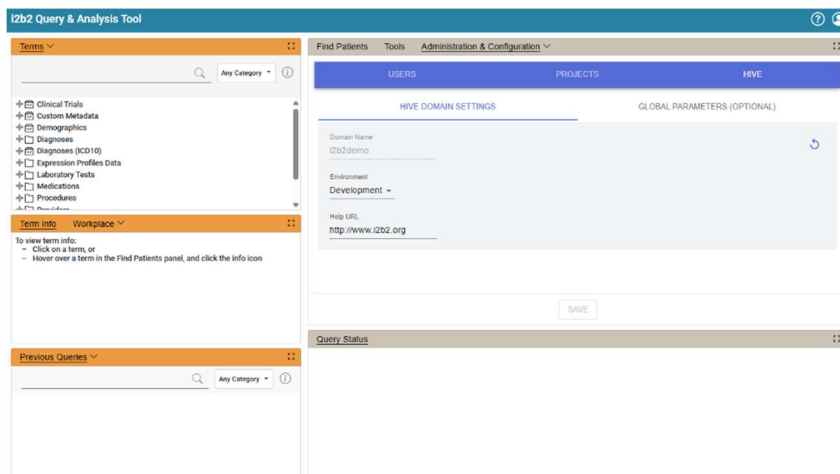
Parameters entered on a project are known as **Hive** or **Global Parameters** and are *specific to the hive (domain)*. A hive parameter is not specific to any one cell, project or user.

4.2.1 Add a Hive (Global) Parameter

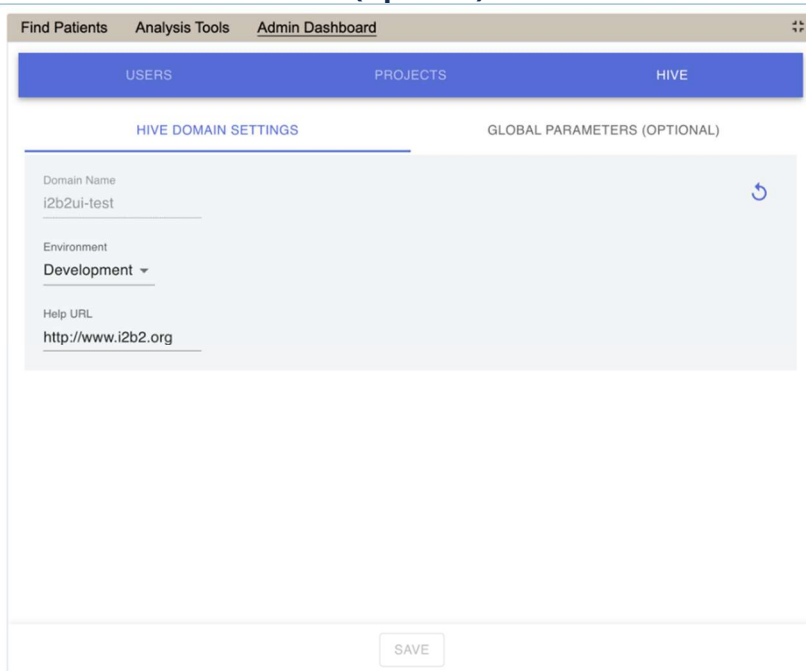
The following steps outline the process of adding a parameter to the hive from within the i2b2 Administration Module.

Assumption: The following steps assume you are already logged into the i2b2 Administration Module. If you do not know how to login please see the chapter titled *Administration & Configuration Plugin Introduction*.

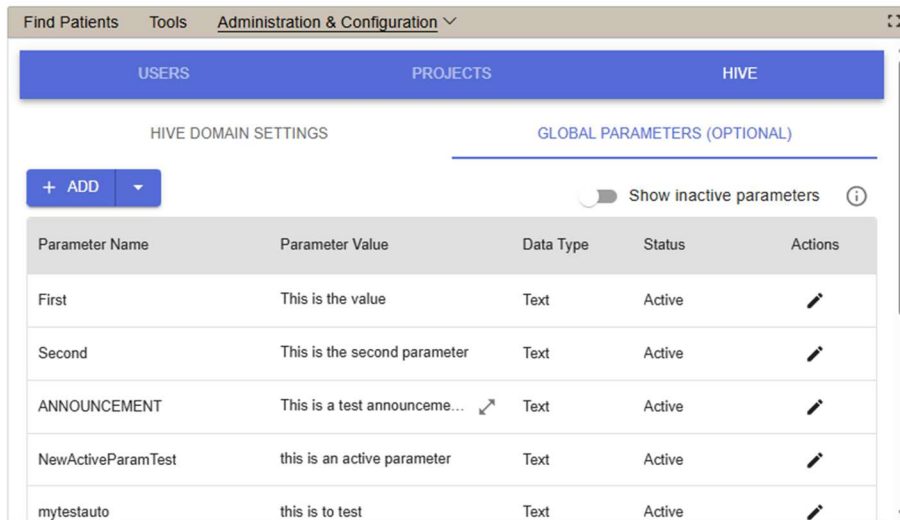
1. In the Admin Dashboard, click on **Hive**.



2. Click on **Global Parameters (Optional)**.



3. The **Parameters List** page will display.



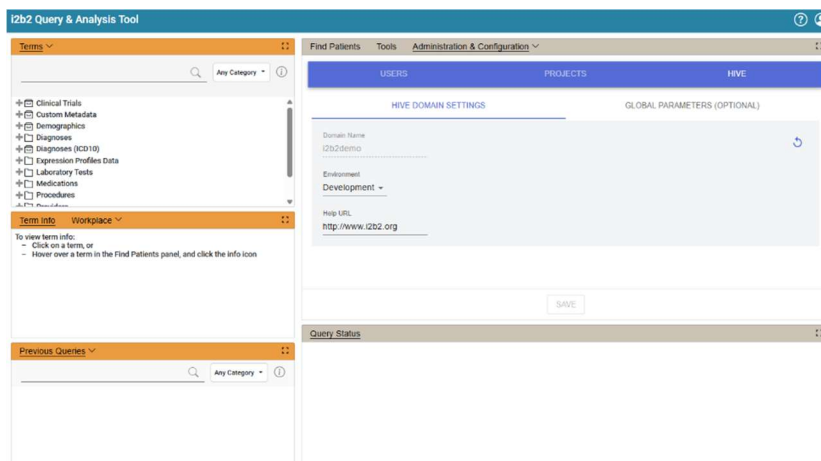
4. Click on **Add**. The *Enter Parameter* page will display.
5. Enter the name of the parameter, the value for the parameter and the data type for the parameter.
6. Click on **Save** to save the new parameter.
7. The *Parameters List* page will display with the new parameter.

4.2.2 Edit a Hive (Global) Parameter

The following steps outline the process of editing a hive parameter in the i2b2 Administration Module.

Assumption: The following steps assume you are already logged into the i2b2 Administration Module. If you do not know how to login please see the chapter titled *Administration & Configuration Plugin Introduction*.

1. In the Admin Dashboard, click on **Hive**.



2. Click on **Global Parameters (Optional)**.

The screenshot shows the Admin Dashboard with the 'Global Parameters (Optional)' section active. The 'HIVE DOMAIN SETTINGS' section is also visible. The 'Global Parameters (Optional)' section contains a 'Domain Name' field with the value 'i2b2ui-test', an 'Environment' dropdown menu set to 'Development', and a 'Help URL' field with the value 'http://www.i2b2.org'. A 'SAVE' button is located at the bottom right of the form.

3. The **Parameters List** page will display.

The screenshot shows the 'Parameters List' page. It features a table with columns: Parameter Name, Parameter Value, Data Type, Status, and Actions. The table contains five rows of parameters. A '+ ADD' button is located at the top left of the table, and a 'Show inactive parameters' toggle is at the top right. The 'Actions' column contains pencil icons for each row, indicating that parameters can be edited.

Parameter Name	Parameter Value	Data Type	Status	Actions
First	This is the value	Text	Active	
Second	This is the second parameter	Text	Active	
ANNOUNCEMENT	This is a test announceme...	Text	Active	
NewActiveParamTest	this is an active parameter	Text	Active	
mytestauto	this is to test	Text	Active	

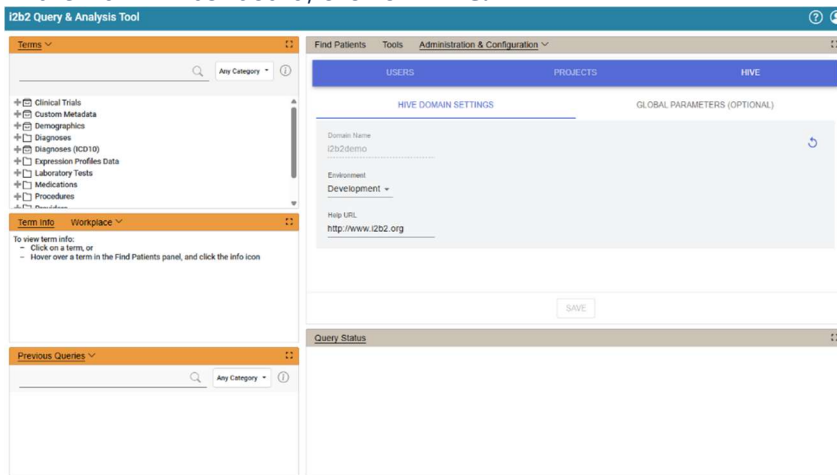
4. Click the **pencil icon** in the row of the **name of the parameter** you want to edit.
5. The *Edit Parameter* fields will display for that row.
6. Make the changes to the parameter and click on the **Save** icon to save the changes.

4.2.2 Delete a Hive (Global) Parameter

The following steps outline the process of deleting a hive parameter in the i2b2 Administration Module.

Assumption: The following steps assume you are already logged into the i2b2 Administration Module. If you do not know how to login please see the chapter titled *Administration & Configuration Plugin Introduction*.

1. In the Admin Dashboard, click on **Hive**.



2. Click on **Global Parameters (Optional)**.

The screenshot shows the 'Admin Dashboard' with tabs for 'Find Patients', 'Analysis Tools', and 'Admin Dashboard'. Below the tabs are three main sections: 'USERS', 'PROJECTS', and 'HIVE'. The 'HIVE' section is active, showing 'HIVE DOMAIN SETTINGS' and 'GLOBAL PARAMETERS (OPTIONAL)'. The 'HIVE DOMAIN SETTINGS' form includes fields for 'Domain Name' (i2b2ui-test), 'Environment' (Development), and 'Help URL' (http://www.i2b2.org). A 'SAVE' button is at the bottom.

3. The **Parameters List** page will display.

The screenshot shows the 'Administration & Configuration' page with tabs for 'Find Patients', 'Tools', and 'Administration & Configuration'. Below the tabs are three main sections: 'USERS', 'PROJECTS', and 'HIVE'. The 'HIVE' section is active, showing 'HIVE DOMAIN SETTINGS' and 'GLOBAL PARAMETERS (OPTIONAL)'. The 'GLOBAL PARAMETERS (OPTIONAL)' section has a '+ ADD' button and a 'Show inactive parameters' toggle. Below is a table of parameters.

Parameter Name	Parameter Value	Data Type	Status	Actions
First	This is the value	Text	Active	
Second	This is the second parameter	Text	Active	
ANNOUNCEMENT	This is a test announceme...	Text	Active	
NewActiveParamTest	this is an active parameter	Text	Active	
mytestauto	this is to test	Text	Active	

4. Click the **pencil icon** in the row of the **name of the parameter** you want to delete.
5. The *Edit Parameter* fields will display for that row.
6. Select **Deleted** from the **Status field** dropdown in the row of the **name of the parameter** you want to delete.

Find Patients
Tools
Administration & Configuration

USERS
PROJECTS
HIVE

HIVE DOMAIN SETTINGS
GLOBAL PARAMETERS (OPTIONAL)

+ ADD
Show inactive parameters

Parameter Name	Parameter Value	Data Type	Status	Actions
First	This is the value	Text	Active	
Second	This is the second parameter	Text	Active	
ANNOUNCEMENT	This is a test announcement This is a...	Text	Hidden	
NewActiveParamTest	this is an active parameter	Text	Deleted	
mytestauto	this is to test	Text	Active	

- Confirm the action. The parameter will be removed from the list of hive (global) parameters on the page, but can still be displayed by toggling on "Show inactive parameters".

5. Activity & Usage Monitoring Plugin Introduction

As an i2b2 Administrator, you can use the *Activity & Usage Monitoring* Plugin to monitor user activity and help troubleshoot user issues when needed. The following chapters outline the information provided in this plugin.

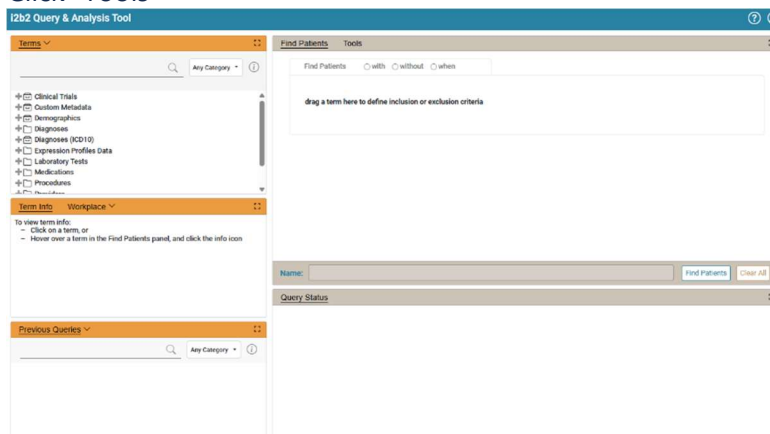
The main goals of the module are to help:

1. Monitor User Activity
2. Support reporting needs related to user activity
3. Facilitate troubleshooting when user queries are having issues

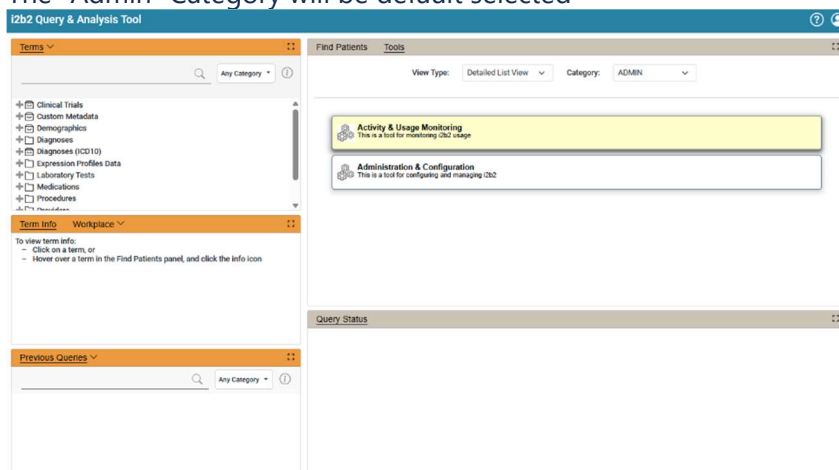
5.1 Accessing Activity & Usage Monitoring Module

The i2b2 Activity & Usage Monitoring Module can be accessed as a plug in.

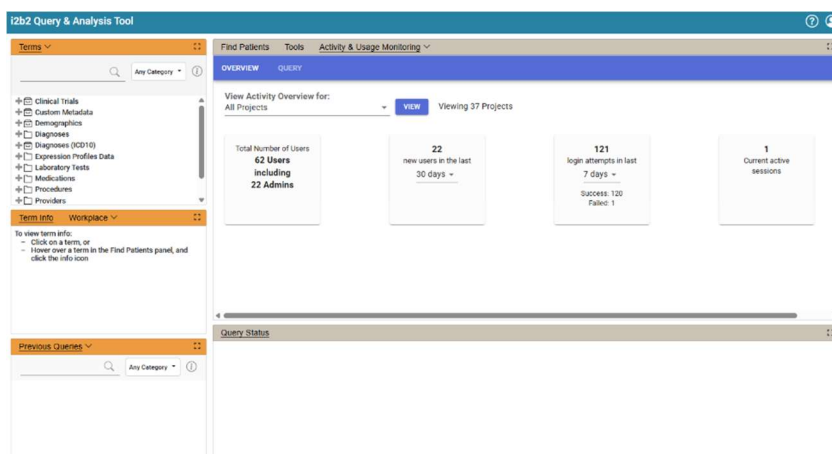
1. Click “Tools”



5. The “Admin” Category will be default selected



2. Click “Activity & Usage Monitoring”



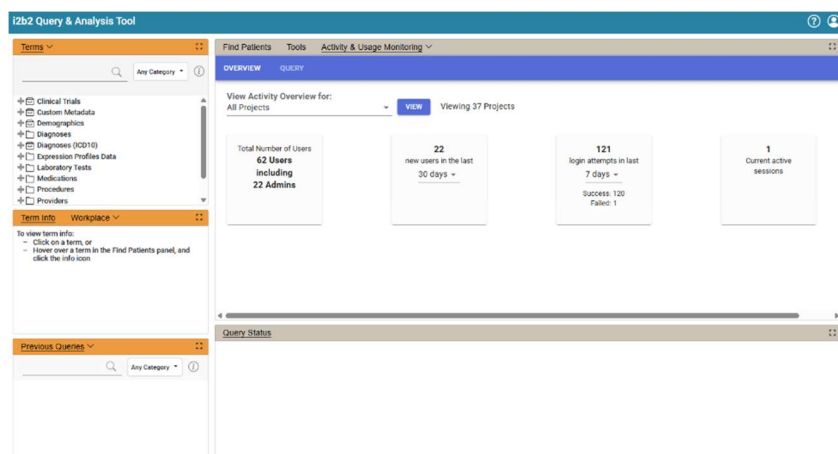
3. This will load the i2b2 Activity & Usage Monitoring Module displaying the Overview tab.

6. Overview of User Activity

The Overview tab in Activity & Usage Monitoring Module enables an Admin to view key metrics and information for an i2b2 instance. You can view usage for all projects in a data source, or view usage details for any specified project.

6.1 Overview for All Projects

To view usage information for all projects in a data source, make sure **All Projects** is selected, and then hit **View**. If your i2b2 instance only has one data source, then usage information for all projects will default display when you open the Activity & Usage Monitoring Module.



The Activity Overview for all projects includes 4 tiles summarizing key information:

1. Total Number of Users

- 1.1. This tile shows a count of the distinct users associated with at least 1 project, as well as a count of distinct Admins.

2. New Users

- 2.1. The tile displays a count of new user-project associations (for distinct users) within a specified time frame.
- 2.2. The admin can select a period of time via drop-down, and the count will automatically update.
- 2.3. NOTE: a newly created Admin will be counted as a new user even if not associated to any projects

3. Login Attempts

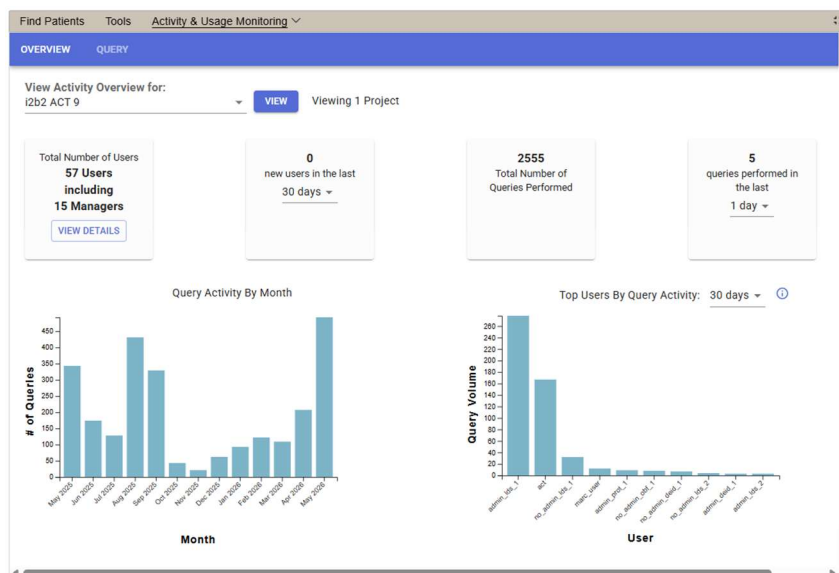
- 3.1. The tile displays a count of login attempts within a specified time frame, with a breakdown of successful vs unsuccessful attempts.
- 3.2. The admin can select a period of time via drop-down, and the count will automatically update.
- 3.3. Note: Failed attempts to change a password are included in the failed login count

4. Current Active Sessions

- 4.1. Shows the number of active sessions across all projects. (This does not update automatically if new sessions begin while admin is on overview tab.)

6.2 Overview for Individual Projects

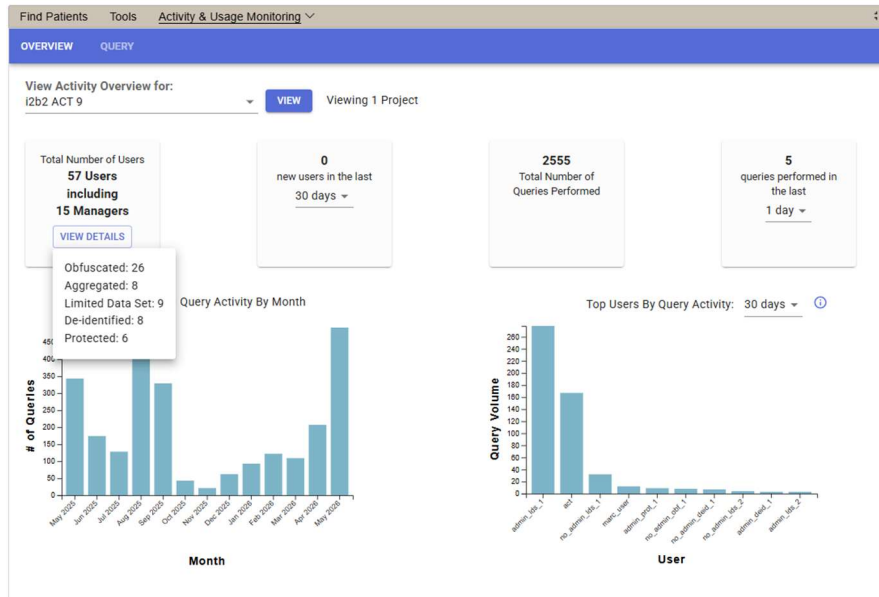
To view usage information for an individual project, enter the project name or select it from the drop down, and this hit **View**. If your i2b2 instance has multiple data sources, then you will first need to select the appropriate data source.



The Activity Overview for an individual project includes 4 tiles and 2 bar charts summarizing key information:

1. Total Number of Users

- 1.1. This tile shows a count of the distinct users associated the specified project, as well as a count of distinct Admins.
- 1.2. Click **View Details** to see the breakdown counts of users by data role.



2. New Users

- 2.1. The tile displays a count of users who have been newly associated with the specified project within a selected time frame.
- 2.2. The admin can select a period of time via drop-down, and the count will automatically update.

3. Total Number of Queries

- 3.1. This tile displays a count of the total number of queries performed by users in the specified project.

4. Recent Queries

- 4.1. The tile displays a count of queries performed in the project within a selected time frame.
- 4.2. The admin can select a period of time via drop-down, and the count will automatically update.

5. Bar Chart: Query Activity By Month

- 5.1. This bar chart show the total number of queries performed per month over the past year.
- 5.2. Hover over a specific bar or month to see the exact query count for that month

6. Bar Chart: Top Users By Query Activity

- 6.1. This chart shows the users that have performed the most queries in the selected project for a specified period of time.
- 6.2. The admin can select a period of time via drop-down, and the chart will automatically update.
- 6.3. The chart will show up to 10 users. If less than 10 users have performed queries in a project, the chart will only reflect users who have performed at least 1 query in the specified time frame.

6.4. Hover over a specific bar or user ID to see the exact query count

7. Query Activity

The Query tab in Activity & Usage Monitoring Module enables an Admin to view detailed data for all recent queries. You can view queries for all projects in a data source, or view query details for any specified project.

7.1 Query Data

The **Query Tab** captures the following general information about a user.

Field	Description	Required
Query Master ID	A unique id for the query. Query Master ID is unique per query within a project.	Yes
Query Name	- Name of the Query - The name is initially generated automatically, and can be edited by the user.	Yes
Project	Name of the project where queries have been performed	Yes
User Name	User Name of the user who initiated the query	Yes
Start Date	Timestamp of when the query starts	Yes
Run Time	- How long it takes the query to complete - For queries with a data request, run time is not displayed until data request is complete. Run time will display how long the initial query took to run, not including data requests.	No
Status	- Shows status of the query: Finished: query has finished with no errors Error: query did not finish due to an error Submitted: query included data request and has not yet completed. - Some queries may also show a Processing or Queued status is the initial patient count has not yet returned.	Yes

Field	Description	Required
Data Requests	- Count of data tables requested as part of query - Count will be 0 if no data requests included	No
Count	- Patient Count returned for initial patient count result - Blank for queries still in progress or queries that result in error	No

7.2 Query Activity Table

To view usage information for all projects in a data source, make sure **All Projects** is selected, and then hit **View**. (If your i2b2 instance only multiple data sources, you will first need to select a data source prior to selecting All Projects).

The screenshot shows the 'Activity & Usage Monitoring' tab in a web application. Below the tab, there are filters for 'View Query Activity for: i2b2synthes8.dbo', 'for All Projects', 'last 30 days', and an 'include deleted queries' checkbox. A 'VIEW' button is present. Below the filters is a table with the following columns: Query Master ID, Query Name, Project, User Name, Start Date, Run Time, Status, Data Requests, Count, and Actions. The table contains 8 rows of query activity data.

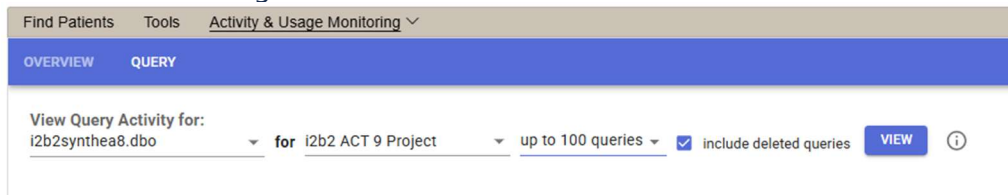
Query Master ID	Query Name	Project	User Name	Start Date	Run Time	Status	Data Requests	Count	Actions
2729	A00-B99 Certain infectious and parasitic diseases (A00-B99)-C00-D49 Neoplasms (C00-D49)@12:3:20	i2b2 ACT 9 Project	no_admin_deid_1	5/28/2026 12:13:42 PM		Submitted	0		
2728	A00-B99 Certain infectious and parasitic diseases (A00-B99)-C00-D49 Neoplasms (C00-D49)@12:3:20	i2b2 ACT 9 Project	no_admin_deid_1	5/28/2026 12:13:39 PM	3.0 secs	Finished	0	2815	
2727	A00-B99 Certain infectious and parasitic diseases (A00-B99)-C00-D49 Neoplasms (C00-D49)@12:3:20	i2b2 ACT 9 Project	no_admin_deid_1	5/28/2026 12:12:34 PM	3.0 secs	Error	1		
2726	A00-B99 Certain infectious and parasitic diseases (A00-B99)-C00-D49 Neoplasms (C00-D49)@12:3:20	i2b2 ACT 9 Project	no_admin_deid_1	5/28/2026 12:12:14 PM	2.9 secs	Finished	0	2815	
2725	A00-B99 Certain infectious and parasitic diseases (A00-B99)-C00-D49 Neoplasms (C00-D49)@12:3:20	i2b2 ACT 9 Project	no_admin_deid_1	5/28/2026 12:03:27 PM		Submitted	2		
2724	Bicarbonate [Moles/volume] in Arterial blood (1960-4)@16:20:28	i2b2 ACT 9 Project	admin_ids_1	5/26/2026 2:47:13 PM	3.1 secs	Error	2		
2723	Bicarbonate [Moles/volume] in Arterial blood (1960-4)@16:20:28	i2b2 ACT 9 Project	admin_ids_1	5/26/2026 11:55:09 AM	4.7 secs	Finished	0	8730	
2722	(t) Lymphocytes (26474-7)-Lymphocytes (26474-7)-	i2b2 ACT 9 Project	admin_ids_1	5/26/2026 11:08:32 AM	2.1 secs	Finished	0	0	

7.2.1 View options for Query Activity Table

When viewing Query Activity, you have the following options:

- Select a specific project or view queries for all project
- Select a set of queries: queries from the last 30, 60, or 90 days, or "up to 100 queries"
 - When a **single project** is selected, "up to 100 queries" the Query Activity table will show the most recent queries up to 100, even if they are not within the last 90 days.

- When **all projects** is selected, the Query Activity table will return an equal number of the most recent queries from each project in the data source.
 - If one or more of the projects have less total queries than the number expected to be returned, then total queries displayed in the Query Activity table will be less than 100.
- **Include deleted queries** displays queries flagged as deleted, which includes queries that a user has deleted from their previous queries window, and queries that a user has canceled after initiating.



7.3 Query Details

The **Actions** column provides a link to show additional query details. When you click on the icon, a new window will appear with 3 tabs, showing the query's XML, SQL, and the query definition. By default the XML tab will display.

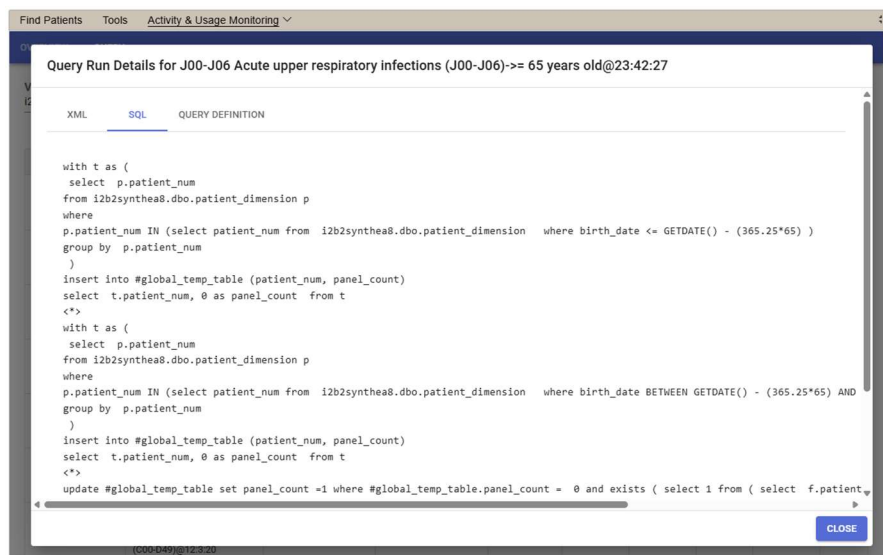
7.3.1 Query XML

The first tab in the Query Details window shows the XML message sent to the server.



7.3.2 Query SQL

To view the SQL query statement, you can select the SQL tab.



7.3.3 Query Definition

The third tab, **Query Definition**, shows the query as it was defined in the web-client, in a similar format to the View Report button in the web-client's Query Status window. If a query has a status of "Finished", the query definition tab will also show the patient count.

